



UNSATURATED POLYESTER RESIN: P-502

Deltech unsaturated polyester grade P-502 is a high quality pre-accelerated, thixotropic, medium viscosity, BS 476: Part-7 approved, Class 2 Fire Retardant resin, primarily developed for use in hand lay, and spray lay, with or without the addition of powdered mineral fillers, especially in medium to large part fabrications.

P-502 exhibits predictable curing characteristics but, like all formulated resin systems, will be affected slightly by variations in ambient conditions, catalyst concentrations, type and concentration of additives such as fillers and pigments etc and normal batch to batch variations during manufacture.

We recommend that the resin be used above 15°C. In cold weather drums should be stored indoors (if possible at least 24 hrs before use). Gel time and curing will be slower at resin temperatures below 15°C and will lengthen as resin temperature decreases.

LIQUID (UNCURED) RESIN DATA

Property	Value	Unit
† VISCOSITY @ 25°C	300 – 450	mPa.s
Thixo Index	2 - 3	
NVM (SOLIDS CONTENT)	60 - 65	%
APPEARANCE	Blue	
SPECIFIC GRAVITY @ 25°C	1.10 – 1.18	
* GEL-TIME @ 20°C,	9 - 14	Minutes
* PEAK EXOTHERM	140 - 170	°C
STABILITY IN DARK @ 25°C ♦	3 (Minimum)	Months

** TYPICAL CURED PROPERTIES

Property	Value	Unit
TENSILE STRENGTH	55	MPa
FLEXURAL STRENGTH	86	MPa
FLEXURAL MODULUS	3.4	GPa
ELONGATION IN TENSION	2.0	%
BARCOL 934-1 HARDNESS	40	
HEAT DISTORTION TEMP	60	°C
WATER ABSORPTION 24 HRS	25	mg

NOTES:

† Using Brookfield LVF viscometer, spindle 2, 60 rpm.

* Measured @ 20°C using Sunshine gel-timer, on the resin 100.0g and 1.0% (w/w) of medium activity MEKP, e.g. Butanox M50 or KP-9, mixed together for 60 seconds. Peak exotherm is measured in the centre of the curing mass.

♦ In unopened original containers.

The information contained in this document is for guidance only and is not intended as a binding specification. Whilst every effort is made to ensure the accuracy of the properties and claimed performance, final selection and use must be warranted by the customer's own suitability proving trials.

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